Evaluation

Introduction

A user interface provides a way of communication between a user and a computer.

Interface design, refers to creating the model of user interfaces for some software.

Before a user interface design is finalized, it needs to be evaluated in terms of quality.

- Evaluation at every stage
- Formative evaluation
  - Early design concepts
  - Different design options
  - Checking for usability problems
- Summative evaluation
  - Finished product

What is evaluation?

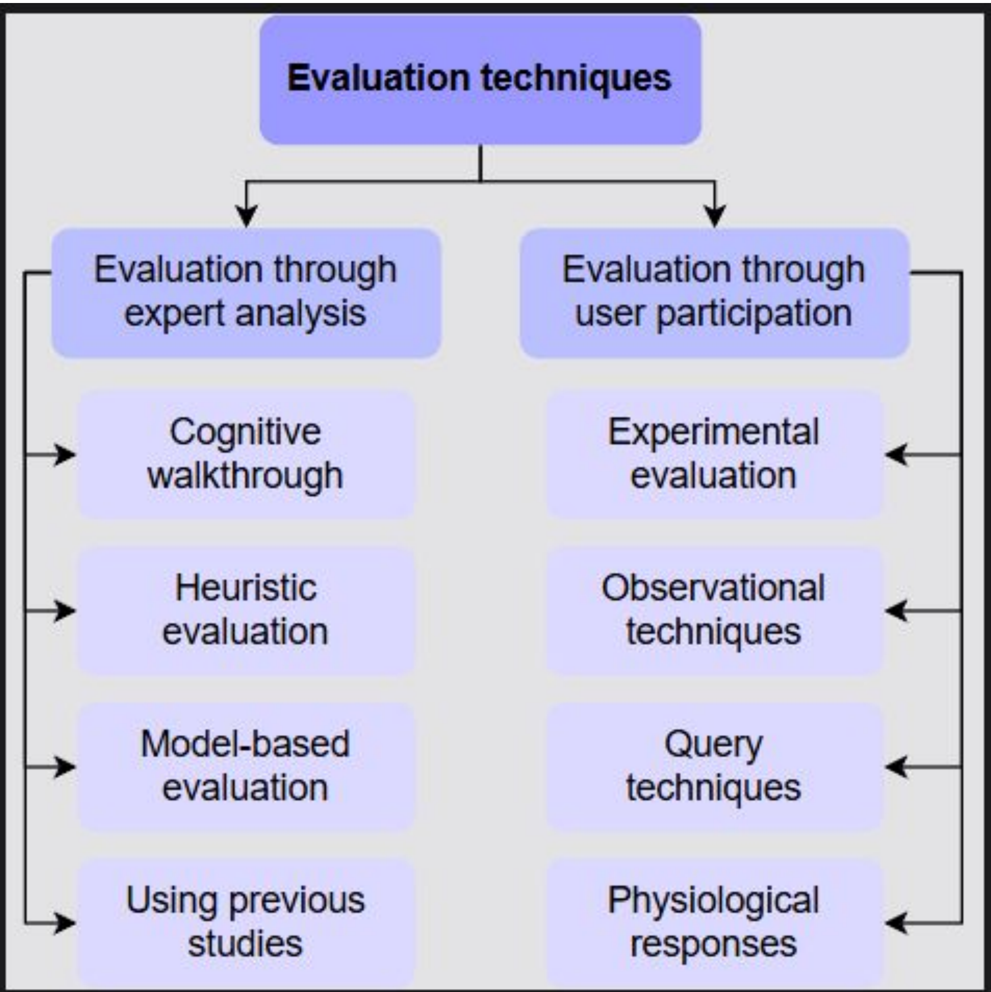
- Evaluation helps test whether the interface meets the set requirements or not.
- A design evaluation <u>highlights problems before the design is sent for the final implementation.</u>
- Ideally, evaluation should not be left to the end of the design process. Instead, it should be a continuous activity where individual interface components are assessed during the design process.

Evaluation techniques

The techniques to perform evaluation can be broadly divided into two categories:

Evaluations that require expert analysis.

Evaluations that require involving user participation.



The evaluation techniques

Evaluation through expert analysis

- Expert analysis is practical when the designer lacks the resources to involve users. Evaluation through inspection by experts can be done through the following methods:
  - **A cognitive walkthrough** involves the evaluators performing the sequence of actions for each task and evaluating their learnability and usability from the users' perspective.
  - In the **heuristic evaluation** technique, evaluators critique the interface design, keeping some usability heuristics or principles in mind. Learn more about heuristic evaluation in this answer.
  - A **model-based evaluation** involves assessing the interface using some models of design specifications. One such model is the GOMS model.
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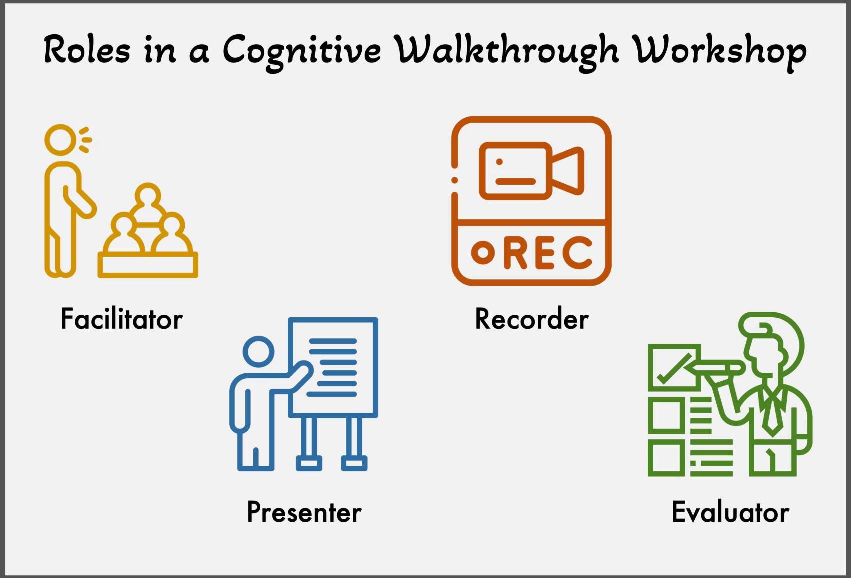
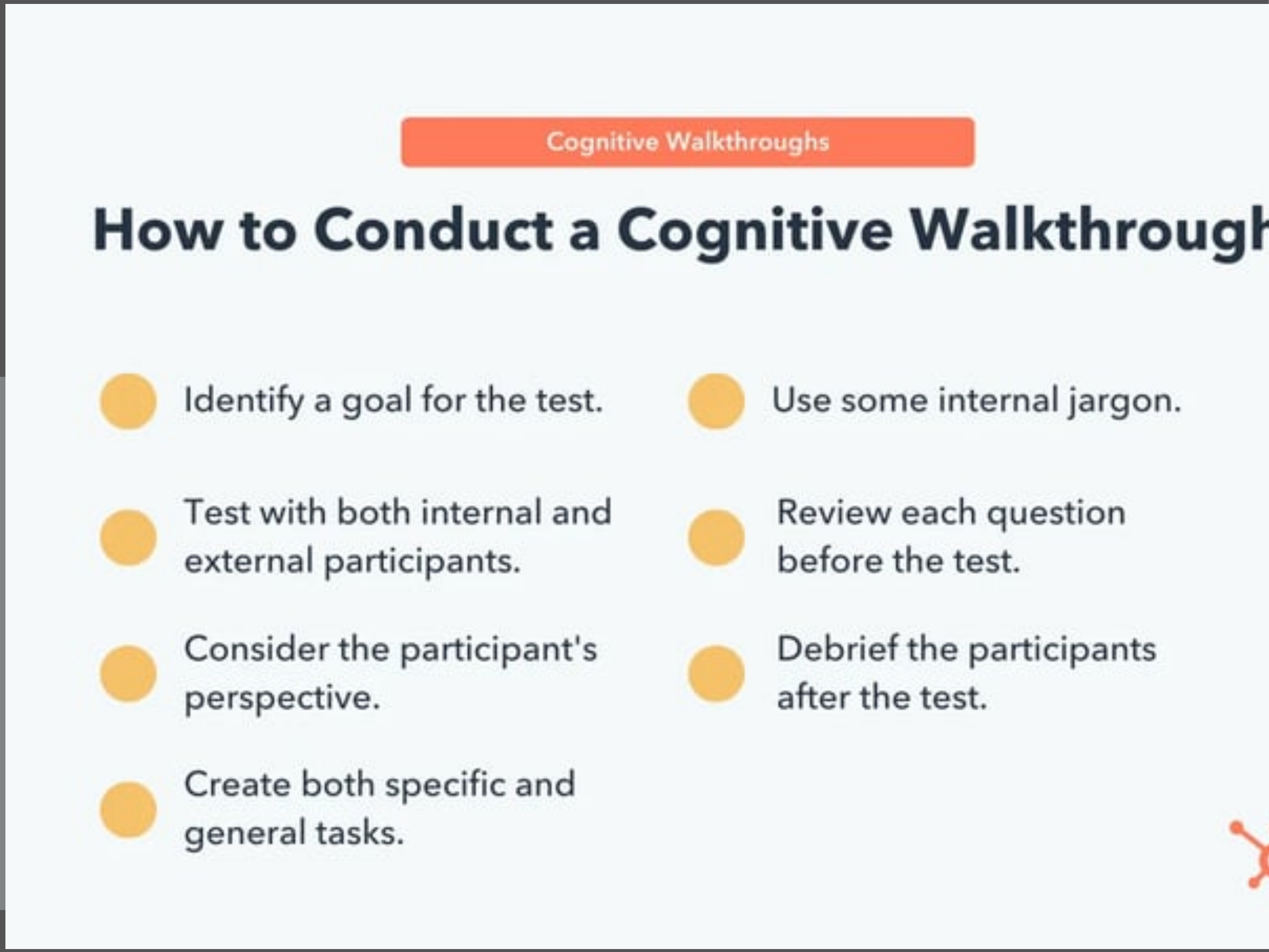

Roles in a Cognitive Walkthrough Workshop
Facilitator
REC
Recorder
Presenter
Evaluator


Cognitive Walkthroughs
How to Conduct a Cognitive Walkthrough
Identify a goal for the test.
Test with both internal and external participants.
Consider the participant's perspective.
Create both specific and general tasks.
Use some internal jargon.
Review each question before the test.
Debrief the participants after the test.


Usability Heuristics
Visibility Of System Status
Match Between System & Real World
User Control And Freedom
Consistency And Standards
STOP
Error Prevention
Ctrl
C
Recognition Rather Than Recall
Flexibility And Efficiency Of Use
Aesthehic And Minimalist Design
Help Users With Errors
Help And Documentation

Evaluation through user participation

- Following are a few evaluation techniques involving participation from actual users.
  - In the **experimental evaluation** technique, controlled experiments are carried out to test specific hypotheses regarding the interface design.
  - The **observational techniques** involve watching the users and recording their behavior to understand their thought processes while using the interface.
  - The **query techniques** include asking the user directly about their experience with the interface, such as interviews and questionnaires.
  - **Monitoring physiological responses** is a way to take physiological measurements of users while they are exposed to the interface and analyze them.

Controlled experiments

- When designer is interested in particular features of a design
- Comparing different designs
- Level of certainty
- Independent VS dependent variables
  - Independent: two different prototypes
  - Dependent: time required to learn
- Confounding variables
  - Learning effects, different background and knowledge

- Within-subject design
  - Same participants in all experiments
- Between-subject design
  - Participants performs in one condition only
- Random sequence
- Interviews and focus groups later on
- Video recording, think aloud

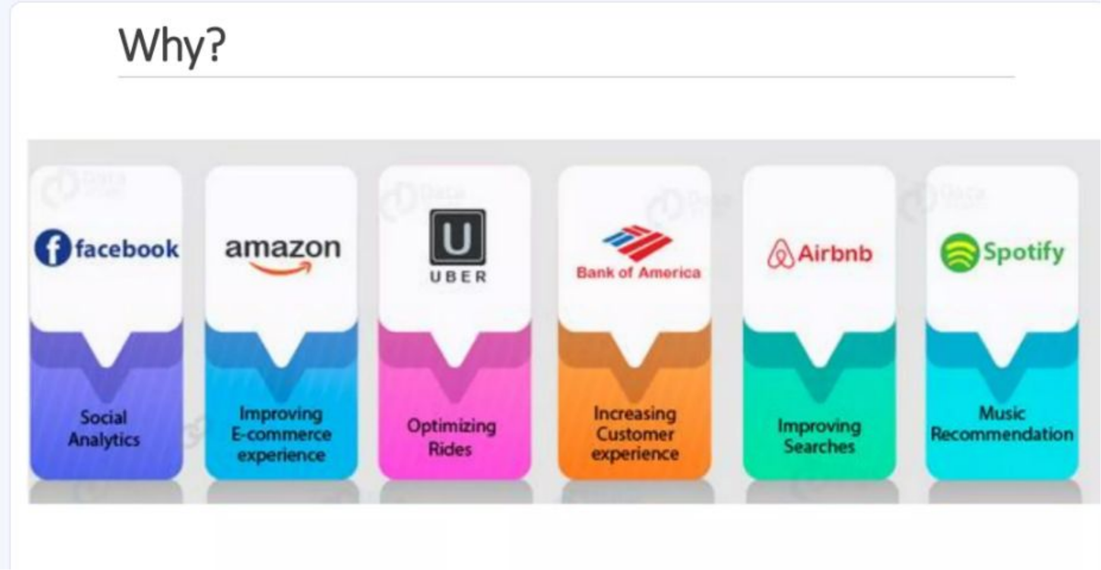
Goals of evaluation

- Assess the system's functionality.
- Assess users' experience with the interface.
- Identify specific problems with the interface

Data analytics

- Era of big data
- Internet of Things
- Data on system performance and user behavior
- Data visualization
- Interesting facts about web and mobile apps
- Sentiment analysis


Understand Customers Behavior
Understand Products Usage
Increase Operational Efficiency
Business Model Innovation


Why?
facebook
Social Analytics
amazon
Improving E-commerce experience
UBER
Optimizing Rides
Bank of America
Increasing Customer experience
Airbnb
Improving Searches
Spotify
Music Recommendation

General Types of Visualizations

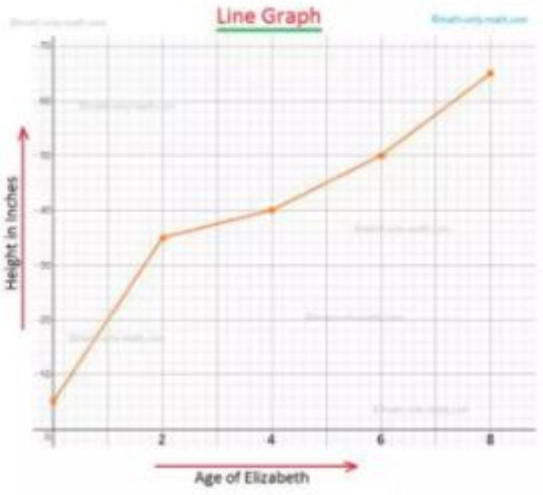
- **Chart:** Information presented in a tabular, graphical form with data displayed along two axes. Can be in the form of a graph, diagram, or map.
- **Table:** A set of figures displayed in rows and columns.
- **Graph:** A diagram of points, lines, segments, curves, or areas that represents certain variables in comparison to each other, usually along two axes at a right angle.
- **Geospatial:** A visualization that shows data in map form using different shapes and colors to show the relationship between pieces of data and specific locations.
- **Infographic:** A combination of visuals and words that represent data. Usually uses charts or diagrams.
- **Dashboards:** A collection of visualizations and data displayed in one place to help with analyzing and presenting data.

Pie Chart

Table

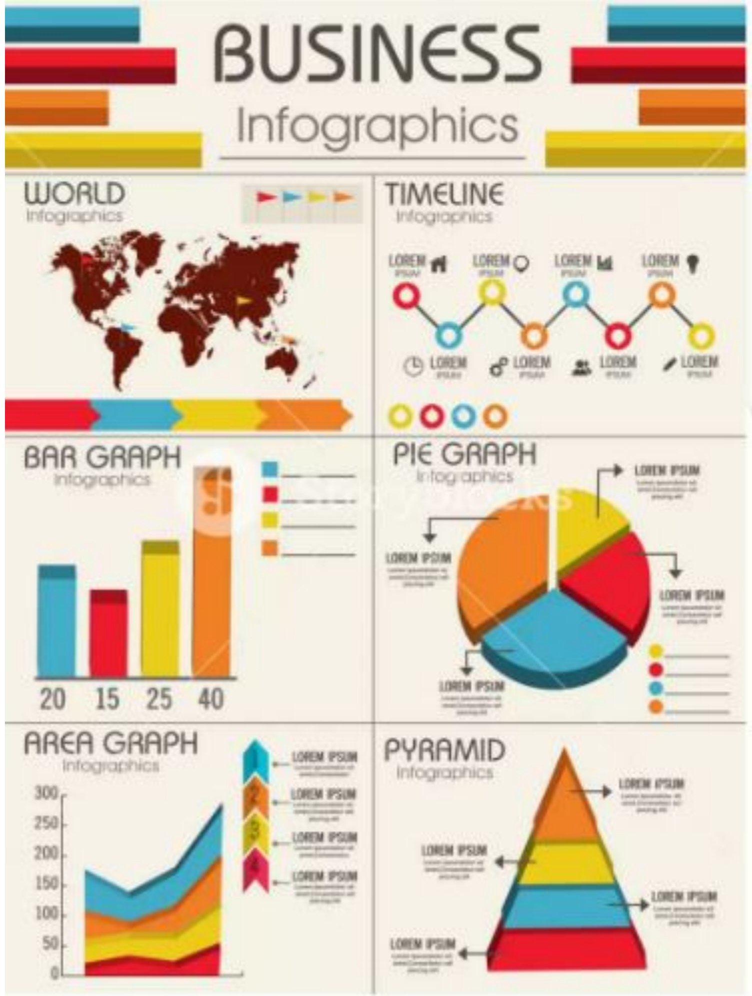
		Column 1	Column 2	
			Column 2.1	Column 2.2
Row 1	Row 1.1			
	Row 1.2			
Row 2				

Table

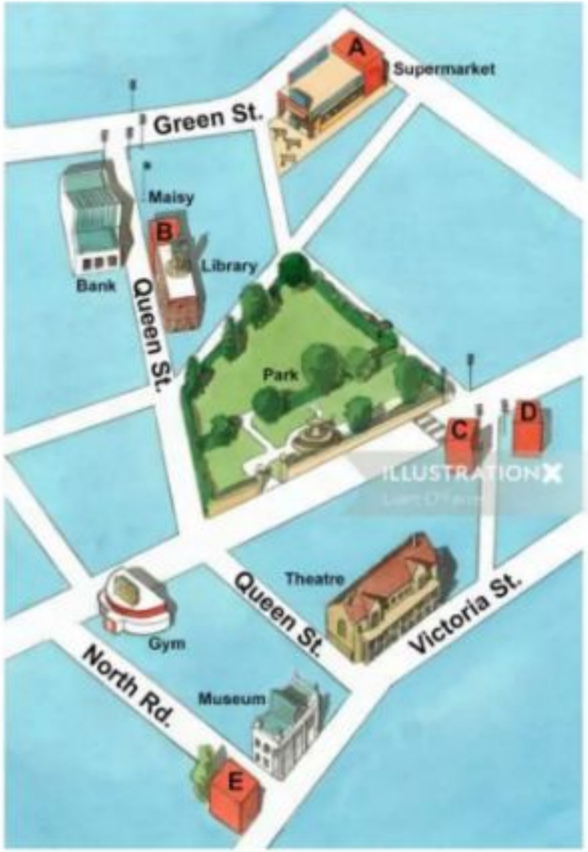


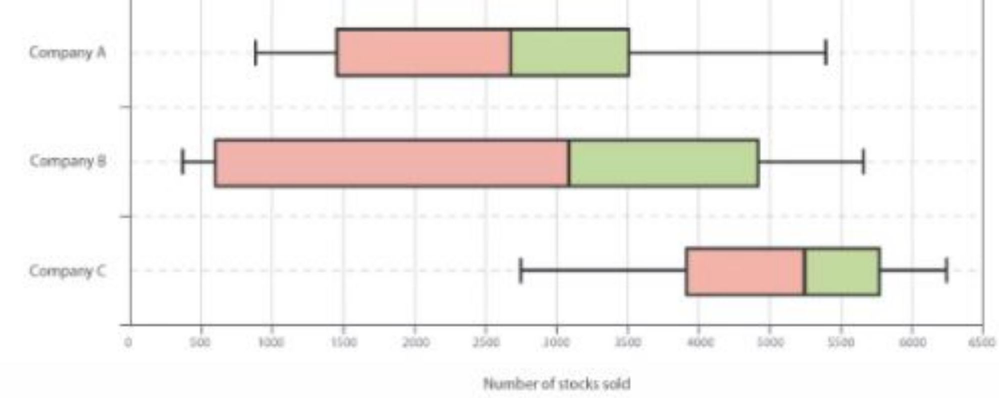
Graph

Geospatial Graph


BUSINESS
Infographics
WORLD
Infographics
TIMELINE
Infographics
BAR GRAPH
Infographics
20 15 25 40
PIE GRAPH
Infographics
AREA GRAPH
Infographics
300
250
200
150
100
50
0
PYRAMID
Infographics
LOREM IPSUM


230
9.86%
$25,690
2,192





Box & Whisker Plot